

Panoramic image → aspect-ratio-preserving input
448 × 448 pixels, checkpoint-specific normalization



EfficientNet-B4
Whole image
ImageNet initialization

DINOv2 ViT-S/14
Whole image
Frozen encoder

DINOv2 ViT-S/14
Whole + 3 regions
Frozen encoder



Global pooled
convolutional features

Whole-image vector
384 features

Whole vector + mean
of regional vectors
768 features



Train encoder +
256-unit head
Weighted cross entropy

Fit standardization +
weighted logistic
regression

Fit standardization +
weighted logistic
regression



Four stage probabilities → most probable reference stage
Evaluation uses held-out predictions, not predictions on fitting images